F18CE2 MV Calculus and Real Analysis B: Tutorial 6

1 State the definition of f is differentiable at $a \in \mathbb{R}$.

Suppose that $f : \mathbb{R} \to \mathbb{R}$ is differentiable at a, and determine the following in terms of $f'(a)$.

$$(a) \quad \lim_{h \to 0} \left[\frac{f(a+4h) - f(a-h)}{h} \right], \qquad (b) \quad \lim_{h \to 0} \left[\frac{f(a+5h) - f(a+2h)}{h} \right].$$

2 Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = |x - 3|$. Prove that f is not differentiable at 3.

3 Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^2, & x \geqslant 0, \\ x^3, & x < 0. \end{cases}$$

Is f differentiable at 0?

4 Let $f : \mathbb{R} \to \mathbb{R}$ be continuous at a. If $g(x) = (x - a)\, f(x)$, prove that g is differentiable at a.

5 Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(x + y) = f(x) + f(y)$ for all x and y in $\mathbb{R}$. Show that $f(0) = 0$. If f is differentiable at 0, prove that f is differentiable at all $a \in \mathbb{R}$.

6 Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(x + y) = f(x)\, f(y)$ for all x and y in $\mathbb{R}$ and $f(0) \neq 0$. Show that $f(0) = 1$. If f is differentiable at 0, prove that f is differentiable at all $a \in \mathbb{R}$.